

Truth-Tracking Profiles: What LLMs Participate In

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Abstract

Debates about large language model grounding often ask whether model representations connect to the world. That framing is too coarse. This paper treats world-connection as a pattern of participation in several stabilising routes to representational success: perception, testimony, coherence, measurement, intervention, error-correction, and practical success. A route is stabilising when it lets facts, or practices already adjusted to facts, constrain what a system produces and how its errors are corrected. The resulting pattern explains why truth-relevant successes travel together, why they fail in patterned ways, and what a system’s stabilising background lets us predict.

Language models are the pressure case. Text-trained systems inherit dense textual traces of testimony and local coherence while weakly participating in perception, measurement, intervention, and practical correction. The result is a familiar failure signature: fluent, coherent outputs can lose answerability when tasks require independent routes to the facts. Self-consistency prompting, retrieval, tools, multimodal input, and feedback should help only where they strengthen the stabiliser the task requires. Surface improvement and truth-tracking improvement can come apart.

I INTRODUCTION: A BINARY PRESUPPOSITION

Do the representations inside a large language model connect to the world? The question is usually put as a yes/no. Bender and Koller (2020) say no: a system trained on form alone, like an octopus eavesdropping on telegraph traffic, learns the statistics of language without touching what language is about. Others reply that such systems meet at least some conditions on reference, or that sensory grounding isn’t required for thought in the first place (Chalmers, 2023; Coelho Mollo & Millière, 2026).

The camps disagree on the verdict but share the question’s shape.

This is the latest form of the symbol-grounding problem (Harnad, 1990): how can symbols, or symbol-like states, get meaning that isn’t merely parasitic on the interpretations supplied by users?

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For LLMs, the same pressure reappears in the question of whether learned linguistic states are grounded in anything beyond linguistic regularity (Pavlick, 2023).

That yes/no framing makes world-connection look like a single capacity, something a system either has or lacks. The successes we call truth-tracking are instead produced by distinct stabilising mechanisms that ordinarily operate together: perception, testimony, coherence, measurement, intervention, error-correction, and practical success. A system can take part in some of these while sitting out the rest. Text-only and weakly tooled language models make that possibility especially visible.

The paper's claim is that those stabilisers form a projectible profile around the representational successes picked out by the predicate *true*. A profile, in this sense, isn't a checklist of individually necessary conditions. It is a pattern of routes whose presence, absence, and coupling let us project how a system will do in nearby cases (Goodman, 1955; Khalidi, 2013).

LLMs matter because they separate routes that human epistemic life normally bundles together. The claim is clearest for systems whose contact with the world remains mediated mainly by human text. Retrieval, tools, multimodal input, and action-guided feedback change that inheritance-heavy profile. The question is which route they add and where improvement should transfer.

The argument is comparative. Self-consistency should help most where the missing support is coherence. Retrieval should help most where the missing support is record access. Preference feedback should improve conversational fit before it improves answerability. Multimodal input should help some perceptual tasks before it helps calibrated measurement. A profile view earns its keep if these contrasts illuminate LLM behaviour better than a single verdict about grounding.

There is also a modest lesson for truth theory. LLMs don't refute correspondence, coherence, pragmatism, reliabilism, deflationism, or social epistemology. They show why each approach captures a real stabiliser once located inside a wider ecology of answerability, correction, and projection. The paper's main task is to make that ecology precise enough to guide comparison.

The argument runs as follows. Section 2 fixes the target: the relevant cases are representational successes (beliefs, assertions, inscriptions, measurements, models, inferential outputs), not truth as an abstract object. Section 3 sets out the stabilising mechanisms and their Boydian lineage. Section 4 says how the profile supports projection under counterfactual perturbation. Section 5 shows how calibration history handles conflict and transfer. Section 6 takes objections. Section 7 develops the language-model profile mechanism by mechanism. Section 8 concludes.

2 THE TARGET: TRUTH-RELEVANT REPRESENTATIONAL SUCCESS

The target has to be fixed before the profile can do any work. Truth, considered as an abstract property or as the role of the truth predicate, isn't the sort of thing that has perceptual, testimonial, or instrumental mechanisms as parts. The target cases are representational successes: beliefs, assertions, inscriptions, measurements, models, and inferential outputs assessed as getting things right.

Some of these are central TRUTHBEARERS. Beliefs and assertions can be true or false in the ordinary sense. Others are neighbouring cases that support truth-ascription by yielding, recording, or constraining claims: a measurement, a map, a statistical model, or an experimental display may be accurate, calibrated, or well-fitted in ways that make some associated propositions true. The profile concerns that broader space of truth-relevant representational success.

That breadth matters because truth-relevant work is often distributed. A thermometer reading, a graph, or a database entry may not be the final assertion, but it can make true assertion possible. A

map can preserve spatial relations well enough to support successful claims about where things are. A model can constrain which inferences remain live. The account is about the background that lets such outputs support getting things right.

This already narrows the ambition. The paper leaves a reductive definition of truth to one side. Deflationary and minimalist theories can insist that the truth predicate has a thin logical or generalising role, and that the basic theory of truth shouldn't be loaded with a substantive causal story (Horwich, 1998). The profile account can grant that point. It asks why some truth-relevant representational successes travel together, support projection, and fail in patterned ways.

The same separation keeps the correspondence intuition in place without making correspondence the whole theory. Correspondence theories preserve the idea that truth involves answerability to how things are (David, 1994; Kirkham, 1992).

For present purposes, ANSWERABILITY names a constraint relation, not a new theory of truth. A representation is answerable to a target when failures in how the target is can bear on the production, uptake, correction, or withdrawal of the representation. A calibrated instrument has answerability because relevant discrepancies can show up in checks, recalibrations, and downstream failures. A coherent but disconnected generator lacks it where no route lets the relevant fact constrain the output. The profile describes how such routes combine and come apart.

The route needn't be immediate. A fact can bear on an output through later uptake, audit, replication, practical failure, or correction of future behaviour. The answerability is stronger when the route is specific, repeatable, and close enough to make the relevant error visible. It is weaker when the route is delayed, generic, or only loosely tied to the claim.

The contrast is practical as well as semantic. If a scale drifts, repeated weighing, standard weights, and failed downstream uses can expose the drift. If a witness is mistaken, records, other witnesses, and later events can defeat the report. If a model generates a plausible citation from textual pattern alone, none of those checks has yet constrained the output. It may still be right, but its rightness is less supported by the route that produced it.

The same sentence can therefore differ in answerability. "The sample contains 3.2 mg of lead" is strongly answerable when it is generated by a calibrated assay whose logs, standards, and downstream checks can expose error. The same string is weakly answerable when a text-only model produces it because such numbers fit the genre. The truth conditions haven't changed. The route by which the claim became available has.

Pragmatist theories mark a different stabiliser. Inquiry, action, and successful practice can correct representations that are locally coherent or socially inherited but poorly connected to the world (Dewey, 1941; James, 1907). That doesn't make truth identical with what works. It says that practical success is one way representational systems become answerable to what they are about.

The explanandum is the stabilising background that makes truth-ascription, accuracy-ascription, and truth-relevant representational success projectible across cases. We ask why some outputs fall inside the extension, why nearby outputs fall outside it, and which features let us project from a partial description of a case to its likely epistemic behaviour. This is where Goodman's projectibility problem enters the paper: the task shifts from attaching the word *true* case by case to finding which patterns make truth-relevant classifications reliable guides to further expectation.

The phrase "truth-tracking" should be read in that modest sense. It doesn't name a hidden essence shared by every true belief, accurate map, and well-calibrated measurement. It names the pat-

terned support that lets us expect some representational successes to survive checks, perturbations, and neighbouring applications better than others.

Boyd's accommodationism supplies the background for that move. Reference and truth aren't secured by definitions in isolation; they are achievements of practices that align perceptual, instrumental, cognitive, and representational activity with relevant causal structures (Boyd, 2021). The present paper accepts that frame and asks a narrower question: how strongly does a given system participate in the practices and mechanisms that make such alignment reliable?

For language models, the lesson is simple. Their outputs are assessed through some stabilisers and not others. A text-only model may inherit textual traces of testimony, statistical regularities, and local coherence pressures while lacking perception, measurement, intervention, and ordinary practical correction. That profile is enough to make truth-relevant assessment possible, uneven, and empirically diagnostic.

3 TRUTH-TRACKING AS A PROJECTIBLE MECHANISM PROFILE

Goodman's new riddle fixes the projectibility problem: the hard question isn't whether past cases support some generalisation, but which predicates support projection from known to unknown cases (Goodman, 1955). The present paper asks that question about truth-relevant representational successes: which stabiliser patterns let us project from partial evidence to likely success or failure?

Khalidi's account supplies the network discipline behind this profile talk. Natural categories earn their status by supporting prediction and explanation, and they do so when associated properties stand in causal relations that make some properties projectible from others (Khalidi, 2013).

The present use is an adaptation. The target cases are the representational successes fixed in Section 2; the stabilisers explain why those cases support projection. Perception, testimony, coherence, measurement, intervention, error-correction, and practical success are linked routes through which representational outputs become more or less answerable to the world. Khalidi's later "nodes in causal networks" formulation gives the graph vocabulary for this representation (Khalidi, 2018).

Calling the result a profile marks two points. First, different routes can dominate in different domains. Proof norms matter more in mathematics than ordinary perception does; calibration matters more in laboratory measurement than conversational fluency does. Second, the epistemic force often lies in the relations among routes. Testimony matters differently when it is backed by records, instruments, and correction than when it is an isolated report.

This profile view is Boydian in lineage. Homeostatic property cluster (HPC) theory explains how real kinds can be projectible without essences (Boyd, 1991, 1999). Boyd's later accommodationism supplies the semantic background: successful reference is an achievement of whole practices, not a relation fixed term by term by accepted definitions. Perceptual skills, instruments, inferential routines, social arrangements, and representational uses all contribute when they help a practice align with relevant causal structures (Boyd, 2021).

Accommodation, in this sense, is the gradual fitting of representational practice to the structures it is used to track. A scientific term, an instrument reading, or an ordinary object name becomes more reliable as perception, measurement, inference, correction, and use are adjusted together. The fit is not secured by a definition alone; it is maintained by the practices that make error detectable and revision possible.

The present claim is narrower than Boyd's. It asks how the accommodation profile changes when

the participant is a text-trained system. An LLM can inherit stabilized linguistic and testimonial patterns while missing the instrumental, perceptual, and practical routes that normally keep those patterns answerable. Stable or projectible profiles shouldn't automatically be called homeostatic; that stronger description should be reserved for cases where corrective coupling has been shown (Reynolds, 2026).

That means the paper is continuous with accommodationism at the explanatory level and diagnostic at the level of system profile. Accommodationism tells us why representational practices can be answerable to causal structure; the profile account tracks how strongly a given system participates in the stabilisers that make that answerability possible. LLMs are the pressure case because they inherit accommodation through human text while missing or weakening several world-corrective paths.

Boyd's homeostatic property cluster is the important special case in which links are actively maintained, where mechanisms correct one another so that the properties co-occur far more reliably than chance. The truth-tracking profile has tightly coupled regions, and the connections thin toward its edges. The profile view doesn't need the whole field to be literally homeostatic before it can project.

This distinction is important for the paper's restraint. A stable association can be useful without being homeostatic. A set of features can support some projection without being maintained by corrective feedback. Homeostasis is the stronger case, where the links are not just observed but actively sustained by mechanisms that pull the profile back into shape when it drifts.

That picture does two jobs. First, it identifies the tightly coupled core. In everyday perception and basic coordination, perception, action, memory, testimony, and practical failure have repeatedly corrected one another. That coupling explains the familiar language-model failure signature: remove the perceptual and verificational nodes, and representational success should fail in a specific way (Section 7).

Second, it places the loose cases. Toward the edges, in theoretical physics, ethics, and some formal domains, the nodes are real but sparsely linked. There may be proof norms, expert testimony, institutional procedures, and coherence constraints, but fewer shared routes of correction. Reading the profile as a network doesn't by itself fix where its boundaries fall: Onishi and Serpico (2022) show that carving a kind from a causal graph stays relative to the chosen level of abstraction, the interest-relativity Craver (2009) found in mechanism talk.

So the paper fixes the epistemic purpose. Following Khalidi's distinction between epistemic purposes and non-epistemic practical or aesthetic ones, the profile is individuated by representing a target domain well enough for relevant inferential, practical, or corrective success under perturbation (Khalidi, 2013). Relative to that purpose, its connections can thin at the edges without the profile dissolving.

4 WHY THE PROFILE IS PROJECTIBLE

Section 2 fixed the target cases, and Section 3 identified the stabilisers. Projectibility concerns the relation between them. The question is whether a partial description of an output and its stabilising background lets us predict further truth-relevant success or failure.

The guiding purpose is representing a target domain well enough for relevant inferential, practical, or corrective success under perturbation. Perturbation matters because it distinguishes lucky fit from robust answerability. A representation may succeed in the training cases, the familiar enviro-

onment, or the current conversational context. It projects more strongly when it continues to guide action, inference, measurement, and correction after conditions change.

A simple example helps. A weather claim repeated from an old webpage may be true when the weather is stable and the page is recent. It projects weakly once the date, location, or local conditions change. A claim constrained by live measurement, calibrated instruments, and error-reporting practices projects more strongly because nearby changes have routes through which they can affect the representation.

Survival and reproduction are limiting biological cases of this structure, not the general explanation. Hoffman-style interface arguments warn that adaptive success and veridical representation can come apart (Hoffman et al., 2015). Martínez (2019) argues that usefulness can drive representations toward truth under specified conditions. The lesson for this paper is methodological: usefulness isn't itself the truth-making relation. It becomes truth-relevant only when the task, environment, and correction channels make error costly in ways that expose the relevant mismatch.

The same point holds outside biology. In science, law, medicine, engineering, testimony, and model evaluation, the stabilisers are often institutional and technological. Replications, instruments, standards, records, peer challenge, and downstream practical failure belong to truth-relevant assessment. They are among the routes by which a truth-relevant output becomes robust under perturbation.

This is also why “works in practice” is too coarse on its own. A useful classification can succeed because it tracks a real structure, because the environment has been arranged around it, or because a task rewards a proxy. Only the first and some forms of the second support the kind of projection at issue here. The profile has to say which route is doing the stabilising.

The profile supports projection when such routes are coupled. A perceptual judgement supported only by immediate appearance projects weakly if it's easily overturned by lighting, perspective, testimony, or measurement. The same judgement projects more strongly when it's embedded in a practice where perception, action, memory, testimony, and correction have repeatedly calibrated one another.

Projection is therefore graded. The question isn't whether a representation is either fully connected or wholly disconnected. It is how much of the relevant correction background the case carries with it. The more the routes have been mutually calibrated, the more confidently success in one nearby case licenses expectations about another.

The privileged stabiliser can change across domains. Some mathematical and logical cases project through inferential structure with little perceptual input. Some scientific cases project through instrumentation and replication instead of ordinary observation. Some testimonial cases project because the institution has mature roles, records, and correction channels. The projectible unit is the coupled pattern.

This also explains why edge cases are expected. Where stabilisers thin or fail to calibrate one another, truth-relevant classifications should project less strongly. Frontier science, remote history, contested legal facts, and ethical disagreement should be treated as thinner regions of the profile. They may still involve real stabilisers, but the pattern supports weaker expectations about agreement, correction, and future success.

The language-model diagnosis follows directly. A system that strongly inherits textual traces of testimony and local coherence, while weakly participating in perception, measurement, intervention,

and practical correction, should show a specific failure profile. It can produce fluent outputs that look truth-relevant from inside the textual profile while failing under perturbations that require independent contact with the world.

Tool use, retrieval, multimodal input, expert review, and action-guided feedback matter because they change the stabiliser pattern. They may strengthen projectibility in some domains, leave it weak in others, or introduce new routes to error. The profile view predicts comparative differences among systems and tasks, not a single verdict about whether language models are grounded or ungrounded.

5 WHAT THE PROFILE STRUCTURE DOES THAT COMPONENT THEORIES DON'T

This section makes good on the claim from Section 1 that the profile view earns its keep. The useful test is whether profile structure answers questions that are hard to formulate when one stabiliser is treated as the main unit of assessment. The answer turns on coupling: the evidential fact is sometimes the way mechanisms have calibrated one another, not the bare performance of any one mechanism taken in isolation.

The component theories enter as local successes. Goldman-style reliabilism foregrounds belief-forming processes and their conditional reliability (Goldman, 1979). Bonjour-style coherentism foregrounds inferential and explanatory integration, while also insisting that coherence needs input if it's to be truth-conducive (Bonjour, 1985). Pragmatist accounts stress correction through successful practice (Dewey, 1941; James, 1907). Correspondence accounts preserve answerability to the world (David, 1994; Kirkham, 1992). Social epistemology treats testimony and institutions as genuine epistemic sources (Coady, 1994; Goldberg, 2010; Lackey, 2008).

These claims aren't concessions made for politeness. They are part of the positive view. If there were no real stabilising work done by reliable production, coherence, answerability, action, testimony, or institutions, there would be no profile to study. The question is what happens when those stabilisers are separated, recombined, or moved into a new system.

The profile view begins after those successes are granted. Its unit is the pattern of calibration among stabilisers: perception checked by testimony, testimony disciplined by memory and records, coherence constrained by measurement, measurement corrected by intervention, and practice pressured by failure. Read as a causal network (Section 3), the profile has structure that source-by-source assessment leaves in the background: the strength and history of the links between nodes.

The methodological point is simple. The account doesn't say that each familiar theory is wrong. It says that a source's evidential force can change when the rest of the correction network changes. A locally reliable route may stop projecting when it is removed from the practices that made it reliable.

LLMs sharpen this point for reliabilism in particular. The relevant process is rarely just "the model generated this text". It may include pretraining data, filtering, instruction tuning, RLHF, retrieval, tool calls, prompting, deployment constraints, post-hoc checking, and institutional uptake. Reliability often attaches to that system-profile. A cleanly bounded internal generator may be too narrow. Reliabilism can preserve its central insight by individuating the process at the same level at which the stabilisers actually operate.

A concrete transfer case shows the pressure. Suppose a retrieval-augmented system scores well on source-backed factual questions about clinical guidelines. A coarse process description such as "this system answers medical questions reliably" invites transfer to neighbouring medical tasks. The profile view withholds that transfer. The success was supported by record access, text extraction,

and institutional testimony. It should transfer to nearby record-backed questions, but it shouldn't transfer unchanged to tasks whose missing stabiliser is live measurement, patient-specific observation, calculation, or intervention.

This isn't a refutation of reliabilism. A careful reliabilist can individuate the process more narrowly, as retrieval-backed answering under a particular profile of records, tools, and checks. At that point the reliabilist has reproduced the profile-level carving. The difference is methodological: the profile account requires that carving before the transfer is granted.

5.1 ADJUDICATING MECHANISM CONFLICT

When two stabilisers disagree, say perception reports one thing and well-attested testimony another, a theory focused on one source at a time underdescribes the evidential situation. Goldman-style reliabilism can ask which process type is reliable, including conditional reliability; social reliabilism can extend the assessed process through testimonial production (Goldberg, 2010; Goldman, 1979). Those are genuine resources. The profile question is different: how tightly have these sources been calibrated against one another in this domain?

In Khalidi's terms, causal relations among associated properties underwrite projection; here the relevant projection runs from one stabiliser's output to another's (Khalidi, 2013).

Consider a modest conflict. A subject's visual judgement says that an object has property *P*. A calibrated instrument says that it lacks *P*. Several competent people, drawing on the same instrument tradition, say the instrument is functioning normally. The profile view avoids a standing rule to trust perception, testimony, or the instrument. It asks which sources have historically corrected which others in this region of practice.

In everyday object recognition, ordinary perception, practical feedback, memory, and testimony have long corrected one another. If I seem to see an old friend at the end of a hallway while well-placed testimony rules out her presence, the perceptual judgement can reasonably give way. Coady's discussion of perception, memory, inference, and testimony already pushes in this direction: the sources interpenetrate, and testimony can rationally undermine what presents itself as perception (Coady, 1994).

In a mature instrumental domain, the priority can reverse. Unaided perception may be the weak node, because the tighter profile joins calibrated instruments, records, expert testimony, replication, and repeatable intervention. The same abstract conflict receives different treatment in different domains. Global source reliability is too coarse a contrast. The history of cross-source correction gives one output a stronger projective relation to the rest of the profile.

This makes co-variation epistemically load-bearing. A tightly coupled profile gives a defeasible presumption that a lone divergent output is local error or a boundary case. A loosely coupled profile gives a different presumption: divergence is evidence that the kind of success under assessment projects weakly here. The verdict is still empirical and fallible, but both the pattern of calibration among stabilisers and the assessed reliability of one source enter the evidence.

The point is not that agreement proves truth. Agreement can be manufactured, coerced, or inherited from a shared mistake. The point is that agreement produced by independent, mutually correcting routes has a different evidential meaning from agreement produced by one copied source. The profile account tries to keep that difference visible.

That difference matters when two descriptions agree about local reliability but disagree about transfer. Suppose a model, instrument, or witness has been reliable in a domain where perception,

records, and measurement have repeatedly corrected one another. A source-by-source assessment can preserve that reliability score when the source is moved to a neighbouring domain.

The profile view doesn't automatically preserve it. If the new domain lacks the calibration history that made the earlier success projectible, it predicts weaker transfer even before any present disagreement appears. The divergent prediction is that reliability inherited without its cross-source correction background should degrade under perturbations that remove the missing checks.

5.2 DOMAIN-CLUSTERED CO-VARIATION

The network view also foregrounds a cross-domain pattern that source-by-source accounts tend to leave implicit: disagreement should cluster where the independent marks of connectivity are thin. Khalidi treats causal strictness and network density as dimensions along which categories support stronger or weaker projection (Khalidi, 2013). The stabilisers of truth-relevant success should be densely linked where they have been jointly disciplined, and sparsely linked where they haven't. Persistent disagreement should cluster in the sparsely connected regions.

The prediction is non-circular only if connectivity is characterised without appeal to whether the mechanisms agree, and it can't be read off the graph in a purpose-free way: Onishi and Serpico (2022) show the abstraction level and boundaries stay interest-relative.

The reply fixes the purpose and leaves the interest-relativity visible. Holding the representational aim of Section 4 fixed, connectivity can be characterised by three marks established before present agreement is checked.

First, the stabilisers can share a selection or training history. Perception, naming, action, and memory in ordinary object recognition have been shaped by recurring tasks and recurring error signals. Instruments, calibration routines, and expert reports in mature scientific practices can share a similar history, even when the relevant objects aren't available to unaided perception.

Second, the practice can have institutional age. Age by itself isn't enough; a long-lived mistake remains a mistake. The relevant sign is the accumulation of correction norms, records, role differentiation, and procedures for passing error reports forward. These features make one stabiliser available as a check on another before any present disagreement has been adjudicated.

Third, the domain can have measurement infrastructure. Instruments, standards, replications, and intervention routines let a practice ask whether perception, testimony, or coherence has drifted. Where those infrastructures are thick, the stabilisers can correct one another. Where they are thin, the same stabilisers may continue to operate, but their relations support weaker projection.

These marks are defeasible indicators, not criteria. Disagreement can also be driven by stakes, incentives, ideology, small samples, strategic ambiguity, or value-ladenness. The profile view doesn't predict disagreement from thin connectivity alone. It predicts that disagreement will be especially persistent where those familiar pressures combine with weak cross-source correction.

Once the purpose is fixed, the profile predicts tighter agreement where shared training history, institutional correction, and measurement infrastructure are thick. It predicts looser agreement where those marks are absent. Frontier science, ethics, contested law, and remote history are expected edge cases for that reason: some stabilisers remain real while the links among them thin.

Profile structure adds to the component theories by explaining changes in relative epistemic force. Process reliability, coherence, correspondence, pragmatic success, and testimony remain local successes. Their force changes across domains because the calibration background changes. Section 7

applies that result to language models: systems can inherit textual and coherentist stabilisers while lacking the perceptual and verificational links that usually keep those stabilisers coupled.

6 OBJECTIONS

6.1 INTEREST-RELATIVITY

One objection cuts at the profile structure itself. Onishi and Serpico (2022) argue that Khalidi's causal-network account inherits the interest-relativity Craver (2009) finds in mechanistic accounts: which causal graph you draw, at what level of abstraction, and where you set its boundaries all answer to explanatory interest, so the network never fixes a unique, interest-free carving of kinds. If that's right, the profile of truth-relevant success is only as objective as our purposes.

The reply concedes the premise and limits the consequence. The account makes two claims, both purpose-relative and both revisable: systems missing perceptual and verificational stabilisers should show the failure pattern described in Section 7, and connectivity should track selection history, institutional age, and measurement infrastructure (Section 5). Each holds once the representational purpose is fixed, and either can fail if the stabilisers are misdescribed.

Onishi and Serpico (2022) reach the same general verdict, that interest-relativity is ineliminable, and Khalidi's domain pluralism leaves room for this move (Khalidi, 2013). This account makes the operative interest explicit instead of disowning it.

6.2 RELABELING AND ABSORPTION

A second objection says that the account will be absorbed by the component theories. Reliabilism can already conditionalise process reliability. Social epistemology can extend the assessed process through testimony, records, and institutions. Coherentism can demand input constraints. On this objection, the profile view redescribes familiar epistemology in causal-network language.

The component theories often have resources to mention coupling. The claim is comparative: profile structure makes the calibration history itself the unit from which we project. Section 5 gave the central case. A source can retain its local success record while losing the correction background that made that success travel. The profile view predicts weaker transfer in that case, because the missing cross-source links matter before a fresh reliability score has been established. The divergence is in the default entitlement to transfer, not in what a careful reliabilist could reconstruct after individuating the process at the same grain.

The account shouldn't be read as a replacement for reliabilism, coherentism, correspondence, pragmatism, or social epistemology. It says when their local stabilisers support projection together, when they pull apart, and what failures should follow.

6.3 DERIVATIVE TESTIMONY

A third objection concerns testimony. Testimony isn't just information with a human source. It's an interpersonal epistemic practice involving assertion, responsibility, trust, uptake, and defeaters (Coady, 1994; Goldberg, 2010; Lackey, 2008). If so, giving a language model testimonial standing merely because it was trained on human text misleads. Recent work on AI-testimony presses the same role question from the other direction: conversational systems strain anthropocentric theories of testimony without thereby making them ordinary human testifiers (Freiman, 2024).

The account should accept that point and use the weaker claim. A text-trained model inherits artifacts of testimonial practices: reports, explanations, records, instructions, arguments, and correc-

tions left in text. That inheritance is a genuine stabiliser because many such texts were produced inside practices already partly accommodated to the world.

The stabiliser is also derivative. Training alone doesn't make the model a normal hearer who can trust a speaker under standing norms of assertion. It doesn't make the model a speaker who takes responsibility for what is asserted. That derivative status explains both the reach and the fragility of text-only systems.

6.4 NON-EMPIRICAL DOMAINS

A fourth objection comes from mathematics, logic, ethics, fiction, and law. In these domains, perception, measurement, intervention, and practical success seem peripheral or misplaced. If the profile is built around world-contact, it may overfit ordinary empirical truth.

No single stabiliser is mandatory across the whole profile. In mathematical and logical cases, inferential structure, proof norms, notation, memory, testimony, and error-correction carry much of the projection. In legal and ethical domains, institutional procedures, role differentiation, public reasons, precedents, and contestation can be stabilising even when agreement remains fragile. Fictional truth has its own textual and practice-bound constraints. These are thinner or differently coupled regions, not counterexamples to the profile structure.

6.5 DEFLATIONISM AND PLURALISM

Finally, deflationists and alethic pluralists may object from opposite directions. Deflationists may say that truth has no deep nature for the profile to explain (Horwich, 1998). Pluralists may say that different domains involve different ways of being true, so a single profile is artificial.

Both objections help fix the ambition. The paper isn't analysing the truth predicate or reducing truth to stabilisers. It explains projectible patterns among truth-relevant representational successes. That target is compatible with a thin account of the truth predicate. It also fits domain pluralism, because the profile already allows different stabilisers to be privileged in different regions. The claim concerns the projectible stabiliser profile of truth-relevant success, not truth as one homeostatic kind.

7 THE LLM CASE: PARTICIPATION, NOT A VERDICT

The language-model case starts by fixing the bearer. There isn't one truth-tracking profile shared by everything called an LLM system. A pretrained text model, an instruction-tuned chat model, a retrieval-augmented system, a tool-using agent, a multimodal model, and an experimentally embedded robot have different stabilising backgrounds. The theory asks us to compare them: change the stabilisers, and the expected successes and failures should change too.

The key term is PARTICIPATION. A system participates in a stabiliser to the extent that its outputs are generated, constrained, corrected, or selected by that stabiliser in a way that supports projection to further cases.

Participation can be inherited, as when training text contains the residue of human perception, testimony, measurement, and correction. It can be operational, as when retrieval, tools, sensors, or instruments constrain the present output. It can be corrective, as when feedback changes later behaviour. It can also be merely decorative, as when an output mentions experiments, sources, or measurements without being constrained by any of them.

These distinctions prevent participation from becoming a metaphor. Inherited participation explains why a text-only system can know its way around a domain whose records are dense and stable.

Operational participation explains why retrieval and tools can matter at deployment, not just during training. Corrective participation explains why feedback matters only when it changes future behaviour in ways tied to the relevant error. Decorative association is the failure mode in which the system sounds connected without being constrained.

They also block a circular reading of the account. Participation isn't inferred from success or failure alone. It is individuated by causal and architectural facts that can be checked before the performance test: training objective, data provenance, filtering history, runtime information flow, tool-call logs, sensor access, update channels, and the route by which feedback changes later behaviour. Performance then tests whether those architectural facts were the right stabilisers to mark.

The main interventions can be described in stabiliser terms. Retrieval augmentation queries an external corpus or search system and places retrieved material into the model's context; it adds record access, not observation. Tool use routes a subtask through a calculator, database, code interpreter, API, or instrument; it adds calculation or measurement only when the right tool is called and its result is integrated.

Self-consistency prompting samples multiple completions or reasoning paths and privileges convergence among them; it strengthens internal convergence, not an external check. Instruction tuning and RLHF change the model's policy through demonstrations, preferences, benchmarks, or evaluations; they add a social or evaluative correction channel whose target depends on the feedback. Multimodal input supplies image, audio, video, or sensor information; by itself, it adds perceptual input rather than calibrated measurement.

The test is whether gains follow those routes. Hold the topic, answer format, and apparent difficulty fixed, and vary the missing stabiliser. Retrieval should improve source-backed recall more than novel measurement or action-sensitive causal judgement. Self-consistency should improve coherence-sensitive reasoning more than claims that need a fresh route to the facts. Multimodal input should improve perceptual classification before it improves calibrated measurement. If gains transfer evenly across those contrasts, the profile has carved the stabilisers badly.

A matched clinical test would make this concrete. One item set asks for recommendations recoverable from guidelines or institutional records. A paired set uses the same disease area, answer format, and difficulty cues but requires a patient-specific lab value, observed sign, dose calculation, or intervention-sensitive causal judgement. The profile predicts a retrieval gain on the first set and little transfer to the second unless the needed measurement or tool route is supplied. Retrieval alone producing the same gain on both sets would count against the carving.

This is also a difficulty control. If the second set is merely harder, adding the relevant lab value or calculator shouldn't selectively restore performance. If the missing stabiliser explains the gap, it should.

The table isn't a ranking from bad to good. It is a map of where a system's support should improve and where it should remain fragile. The same deployed product may occupy several rows at once. A chat system can be instruction-tuned, retrieval-augmented, partly tool-using, and multimodal. The profile question asks which stabiliser is live in the task being assessed.

The baseline text-only case is the cleanest pressure test. Such a model inherits textual products of human accommodation: reports written by perceivers, explanations corrected by teachers, records produced by institutions, descriptions shaped by instruments, and genres whose norms often reward coherence and accuracy. That inheritance is why text-only systems can do real epistemic work. It also

System profile	Added or dominant stabiliser	Directional prediction
Text-only pretrained model	Textual inheritance and local coherence	Strong where dense, stable text is a good proxy; fragile for recent, obscure, perceptual, or measurement-dependent facts.
Instruction-tuned or RLHF model	Preference, helpfulness, safety, and benchmark feedback	Better conversational fit and some error reduction; no automatic independent route to the facts.
Retrieval-augmented system	Live access to records and institutional traces	Better on updated or source-backed recall; still weak where the missing route is experiment, perception, or intervention.
Tool-using system	Calculators, databases, code, instruments, or APIs	Better where the right tool is called and interpreted; new failures from tool choice, stale data, and integration errors.
Multimodal system	Perceptual input in image, audio, video, or sensor form	Better for some perceptually anchored tasks; by itself, no calibrated measurement or practical correction.
Action-guided or expert-reviewed system	Practical feedback, review, and downstream failure pressure	Stronger correction profile where feedback is timely and domain-competent; weaker where review is sparse or proxy-based.

Table 1: Architecture-sensitive profile predictions.

explains why the inherited stabiliser shouldn't be called ordinary testimony without qualification. The model has access to testimonial residue, not the standing of a speaker or hearer in a testimonial exchange.

This derivative inheritance has two sides. On the positive side, human text is saturated with world-directed practice. It contains the residue of people looking, measuring, arguing, repairing, citing, and correcting one another. On the negative side, the model doesn't usually re-enter those practices when it produces a fresh answer. It draws on their products without automatically sharing their present checks.

Text-only training can still produce more structure than a bare "textual residue" slogan suggests. Othello-GPT is an important pressure case: a sequence model trained to predict legal moves in a board game developed an internal representation of board state, and interventions on that representation affected its outputs (Li et al., 2023). The profile view should expect such cases where the target structure is densely and systematically encoded in the sequence stream. They aren't counterexamples to the profile; they show that inherited participation can be strong when the relevant world is small, rule-governed, and recoverable from the training signal.

The limit matters. Internal world-model evidence should be strongest where the relevant state variables are encoded in the sequence and the loss function penalises violations of their structure. It should weaken when success depends on variables absent from the sequence, live measurement, chan-

ging local conditions, or intervention in the target system. Robust transfer across those boundaries would count against the profile diagnosis.

Coherence is the second strong stabiliser. Training and decoding reward local well-formedness, genre fit, inferential smoothness, and continuation from context. Those pressures explain why these systems can display impressive formal linguistic competence, in the sense separated from functional world use by Mahowald et al. (2024). They also explain the vulnerability. Coherence can improve the shape of an answer without adding an independent route by which the relevant facts constrain it.

Perception, measurement, intervention, and practical correction are weak in the baseline profile. They enter indirectly through text produced by others, not as live routes by which the current output is checked. A model can generate a plausible lab result, legal summary, local observation, or bibliographic claim because similar text exists. That doesn't mean that an experiment, court record, observation, or catalogue entry has constrained the present output. The profile view locates failures exactly where textual inheritance and coherence cease to be good proxies for the target domain.

This gives hallucination a more specific role. Hallucination has no single cause, and the LLM literature treats it as heterogeneous (Huang et al., 2025). The profile claim is narrower: text-only systems should be especially vulnerable to fluent, coherent, genre-appropriate outputs that fail when the task requires a fresh or independent route to the facts. Those failures should be common for obscure sources, recent events, local observations, novel measurements, fine-grained quantities, and action-sensitive causal claims. They should be less common where dense, stable, redundant text is already a good proxy.

Mitigation results should be read through the same distinction. Retrieval can reduce hallucination in conversation (Shuster et al., 2021), but that is evidence about record access. It doesn't by itself add perception, experiment, or intervention. Tool use and multimodality make parallel changes. They add operational routes where the tool, sensor, or modality is actually live in the task; otherwise the new vocabulary of tools or perception can remain decorative.

Action-guided systems are the strongest possible upgrade in the table, but only when the feedback is timely, attributable, and tied to the relevant failure. A robot that breaks a glass, misses a target, or fails a laboratory protocol can receive correction that a text-only model lacks. A deployed chatbot that receives delayed approval, engagement metrics, or sparse user edits has a thinner corrective route. The label "feedback" hides that difference unless the stabiliser is specified.

Instruction tuning, RLHF, benchmark evaluation, red-teaming, and expert review are correction channels, but they must be classified by what they correct. They often impose helpfulness, harmlessness, preference satisfaction, conversational fit, and benchmark success. Those constraints can reduce some errors, but they can also reward answer-shaped behaviour when users or raters prefer confidence, agreement, or plausible explanation.

Work on sycophancy shows the risk of treating preference feedback as truth feedback without further structure (Shapira et al., 2026). In the profile terms, these processes strengthen some social and evaluative stabilisers while leaving perception, measurement, and intervention proxy-based.

Coelho Mollo and Millière are live interlocutors here. Their vector-grounding argument distinguishes kinds of grounding and argues that some LLM states can satisfy causal-historical or teleosemantic conditions for referential grounding (Coelho Mollo & Millière, 2026). The present account needn't deny that. Its explanandum is broader than referential grounding.

Work on neural word embeddings pushes in a similar direction: learned representations may have

genuine, though limited, contents fixed by the functions they serve inside the system (Mallory, 2026). That is friendly pressure on the baseline text-only case. The reply is not to deny content, but to ask which stabilisers that content participates in.

Referential grounding is one dimension of world-connection. Truth-tracking answerability depends on a wider profile of stabilisers, including correction, measurement, intervention, perception, and practical feedback. An LLM state might be referentially grounded in their sense while the deployed system still has a fragile truth-tracking profile for tasks requiring independent checking.

The negative test matters most. Improvements in one route shouldn't automatically transfer to tasks whose missing stabiliser is different. If reliable perception, measurement, retrieval, tool use, expert correction, or action feedback fails to alter the expected contrast in the direction just specified, the profile has misdescribed the mechanisms. If text-only systems remain mainly dependent on inherited textual accommodation and local coherence, the same vulnerability should persist even as fluency improves.

8 CONCLUSION

The question that opened the paper was whether the representations inside a large language model connect to the world. The answer has been a change in what the question asks. World-connection is better treated as a profile of participation in the mechanisms by which representational outputs become answerable to what they are about.

The central claim has been that truth-relevant representational success is supported by a projectible profile of stabilisers: perception, testimony, coherence, measurement, intervention, error-correction, and practical success. These stabilisers don't define truth. They explain why the cases picked out by successful truth-ascription travel together, fail in patterned ways, and support predictions from partial knowledge of a system's stabilising background. A thin truth predicate can coexist with thick truth-tracking practices. Where the links are actively maintained, the coupling can become homeostatic in Boyd's strong sense; elsewhere the profile can remain projectible without being homeostatic throughout.

Reliabilism, coherentism, correspondence theory, pragmatism, and social epistemology each isolate a real stabiliser. The reason to move beyond them lies in their relations: the history of correction among sources, the density of the causal network in which they operate, and the degree to which one route to success can compensate for or expose the failure of another. The epistemic fact is sometimes carried by the pattern, not by a source considered alone.

That is why the view doesn't need to defeat the component theories. It needs to show how their stabilisers interact. In ordinary cases, the interaction is easy to miss because the routes run together. In language models, the routes come apart.

A text-only model can inherit substantial textual residues of testimony and local patterns of inferential and discursive coherence. It may even acquire rich internal structure where the target domain is densely encoded in the sequence stream. That still differs from ordinary perceptual contact, calibrated measurement, intervention, and correction through action.

The result is comparative, not classificatory. Retrieval, multimodal input, tool use, expert review, experimental embedding, and action-guided feedback alter the stabiliser profile. The question is which route they add, whether that route is live in the task being assessed, and where improvement should transfer. A system can improve along one dimension while remaining fragile along another.

The account is empirical. Participation is fixed by causal and architectural facts before performance is checked, and the predicted contrasts can fail.

The broader lesson extends beyond artificial systems. LLMs are useful pressure cases because they separate mechanisms that human epistemic life usually runs together. They show that linguistic competence, internal coherence, and statistical sensitivity to past assertion don't exhaust truth-tracking. Truth-tracking depends on a wider ecology of correction. Once that ecology is represented as a profile, the interesting question becomes visible: what a system participates in, what it borrows, what it lacks, and where those absences should matter.

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REFERENCES

- Bender, E. M., & Koller, A. (2020). Climbing towards NLU: On meaning, form, and understanding in the age of data. In D. Jurafsky, J. Chai, N. Schluter & J. Tetreault (Eds.), *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 5185–5198). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2020.acl-main.463>
- BonJour, L. (1985). *The structure of empirical knowledge*. Harvard University Press.
- Boyd, R. (1991). Realism, anti-foundationalism and the enthusiasm for natural kinds. *Philosophical Studies*, 61(1–2), 127–148. <https://doi.org/10.1007/BF00385837>
- Boyd, R. (1999). Homeostasis, species, and higher taxa. In R. A. Wilson (Ed.), *Species: New interdisciplinary essays* (pp. 141–186). MIT Press. <https://doi.org/10.7551/mitpress/6396.003.0012>
- Boyd, R. (2021). Rethinking natural kinds, reference and truth: Towards more correspondence with reality, not less. *Synthese*, 198(S12), S2863–S2903. <https://doi.org/10.1007/s11229-019-02138-4>
- Chalmers, D. J. (2023). Does thought require sensory grounding? from pure thinkers to large language models. *Proceedings and Addresses of the American Philosophical Association*, 97, 22–45. <https://philpapers.org/rec/CHADTR>
- Coady, C. A. J. (1994). *Testimony*. Oxford University Press. <https://doi.org/10.1093/0198235518.001.0001>
- Coelho Mollo, D., & Millière, R. (2026). The vector grounding problem. *Philosophy and the Mind Sciences*, 7(1). <https://doi.org/10.33735/phimisci.2026.12307>
- Craver, C. F. (2009). Mechanisms and natural kinds. *Philosophical Psychology*, 22(5), 575–594. <https://doi.org/10.1080/09515080903238930>
- David, M. (1994). *Correspondence and disquotatation: An essay on the nature of truth*. Oxford University Press. <https://doi.org/10.1093/oso/9780195079241.001.0001>
- Dewey, J. (1941). Propositions, warranted assertibility, and truth. *The Journal of Philosophy*, 38(7), 169–186. <https://doi.org/10.2307/2017978>
- Freiman, O. (2024). AI-testimony, conversational AIs and our anthropocentric theory of testimony. *Social Epistemology*, 38(4), 476–490. <https://doi.org/10.1080/02691728.2024.2316622>
- Goldberg, S. C. (2010). *Relying on others: An essay in epistemology*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199593248.001.0001>
- Goldman, A. I. (1979). What is justified belief? In *Justification and knowledge* (pp. 1–23). Springer Netherlands. https://doi.org/10.1007/978-94-009-9493-5_1
- Goodman, N. (1955). *Fact, fiction, and forecast*. Harvard University Press.
- Harnad, S. (1990). The symbol grounding problem. *Physica D: Nonlinear Phenomena*, 42(1–3), 335–346. [https://doi.org/10.1016/0167-2789\(90\)90087-6](https://doi.org/10.1016/0167-2789(90)90087-6)
- Hoffman, D. D., Singh, M., & Prakash, C. (2015). The interface theory of perception. *Psychonomic Bulletin & Review*, 22(6), 1480–1506. <https://doi.org/10.3758/s13423-015-0890-8>
- Horwich, P. (1998). *Truth*. Oxford University Press. <https://doi.org/10.1093/0198752237.001.0001>
- Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B., & Liu, T. (2025). A survey on hallucination in large language models: Principles, taxonomy,

- challenges, and open questions. *ACM Transactions on Information Systems*, 43(2), Article 42, 1–55. <https://doi.org/10.1145/3703155>
- James, W. (1907). *Pragmatism: A new name for some old ways of thinking*. Longmans, Green, Co. <https://www.gutenberg.org/ebooks/5116>
- Khalidi, M. A. (2013). *Natural categories and human kinds: Classification in the natural and social sciences*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511998553>
- Khalidi, M. A. (2018). Natural kinds as nodes in causal networks. *Synthese*, 195(4), 1379–1396. <https://doi.org/10.1007/s11229-015-0841-y>
- Kirkham, R. L. (1992). *Theories of truth: A critical introduction*. MIT Press.
- Lackey, J. (2008). *Learning from words: Testimony as a source of knowledge*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199219162.001.0001>
- Li, K., Hopkins, A. K., Bau, D., Viégas, F. B., Pfister, H., & Wattenberg, M. (2023). Emergent world representations: Exploring a sequence model trained on a synthetic task. *The Eleventh International Conference on Learning Representations*. https://openreview.net/forum?id=DeGo7_TcZvT
- Mahowald, K., Ivanova, A. A., Blank, I. A., Kanwisher, N., Tenenbaum, J. B., & Fedorenko, E. (2024). Dissociating language and thought in large language models. *Trends in Cognitive Sciences*, 28(6), 517–540. <https://doi.org/10.1016/j.tics.2024.01.011>
- Mallory, F. (2026). Teleosemantics for neural word embeddings. *Mind and Language*. <https://doi.org/10.1111/mila.70037>
- Martínez, M. (2019). Usefulness drives representations to truth: A family of counterexamples to hoffman’s interface theory of perception. *Grazer Philosophische Studien*, 96(3), 319–341. <https://doi.org/10.1163/18756735-09603004>
- Onishi, Y., & Serpico, D. (2022). Homeostatic property cluster theory without homeostatic mechanisms: Two recent attempts and their costs. *Journal for General Philosophy of Science*, 53(1), 61–82. <https://doi.org/10.1007/s10838-020-09527-1>
- Pavlick, E. (2023). Symbols and grounding in large language models. *Philosophical Transactions of the Royal Society A*, 381, 20220041. <https://doi.org/10.1098/rsta.2022.0041>
- Reynolds, B. (2026). *Not every stable cluster is homeostatic: Stability, network order, and control in projectible kinds* [Manuscript archived at PhilArchive, 1 June 2026]. <https://philarchive.org/rec/REYNES>
- Shapira, I., Benade, G., & Procaccia, A. D. (2026). How RLHF amplifies sycophancy. *arXiv preprint arXiv:2602.01002*. <https://doi.org/10.48550/arXiv.2602.01002>
- Shuster, K., Poff, S., Chen, M., Kiela, D., & Weston, J. (2021). Retrieval augmentation reduces hallucination in conversation. *Findings of the Association for Computational Linguistics: EMNLP 2021*, 3784–3803. <https://doi.org/10.18653/v1/2021.findings-emnlp.320>